

KARTHIK D

ASPIRING DATA SCIENTIST

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SUMMARY

Aspiring Data Scientist with hands-on experience in **Python, SQL, statistics, data analysis, and machine learning** through practical projects. Skilled in **data cleaning, exploratory data analysis, preprocessing, model development, evaluation, and basic ML deployment**. Seeking a Data Science or Machine Learning Internship to apply and strengthen practical skills in real-world projects.

EDUCATION

B.Tech – Information Technology 2022 – 2026
AVS College Of Technology, Tamil Nadu
CGPA: 8.5/10
Higher Secondary Certificate (HSC) 2022
St.pauls Higher Secondary School, Tamil Nadu

PROJECTS

SEMICONDUCTOR MANUFACTURING QUALITY PREDICTION Sep 2026

Tools: Python, Pandas, NumPy, Scikit-learn, Machine Learning, SMOTE

- **Developed** a binary classification model to predict semiconductor manufacturing quality outcomes (Pass/Fail) using process and sensor features.
- **Performed** data preprocessing, feature selection, scaling, and class-imbalance handling using SMOTE and class-weighted learning.
- **Compared** multiple classification approaches and evaluated models using classification metrics and validation techniques.
- **Built** a production-ready preprocessing and prediction pipeline using SelectKBest, StandardScaler, and Logistic Regression, with probability threshold tuning for classification decisions.

Gold Price Prediction Aug 2026

Tools: Python, Pandas, NumPy, Matplotlib, Scikit-learn, Machine Learning, Gradio

- **Developed** a machine learning regression model to predict 22K gold price per gram in INR based on the USD/INR exchange rate using historical financial data.
- **Performed** data preprocessing and exploratory analysis using Pandas and NumPy to understand the relationship between currency exchange rates and gold prices.
- **Built and evaluated** a regression pipeline with feature scaling and a trained Scikit-learn model, saving the scaler and model for reusable predictions.
- **Deployed** the trained model as an interactive Gradio web application, enabling users to enter the USD/INR exchange rate and obtain the predicted gold price in INR.

E-COMMERCE SALES ANALYSIS July 2026

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, SQL

- **Analyzed** e-commerce transaction data using Python and SQL to identify sales trends, customer behavior, and key business metrics.
- **Cleaned and transformed** raw transactional data by handling missing values, duplicates, inconsistent records, and data types.
- **Performed** exploratory data analysis and created visualizations using Matplotlib and Seaborn to identify product and customer trends.
- **Generated** actionable insights from sales analysis to support data-driven business understanding.

SKILLS

Programming: Python, SQL

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Feature Engineering, Cross-Validation, Model Evaluation

Data Science: EDA, Data Cleaning, Data Preprocessing, Statistical Analysis

Statistics: Descriptive Statistics, Probability, Hypothesis Testing, Correlation

Deployment/Tools: FastAPI, Gradio, Jupyter Notebook, Google Colab, VS Code

Version Control: Git, GitHub